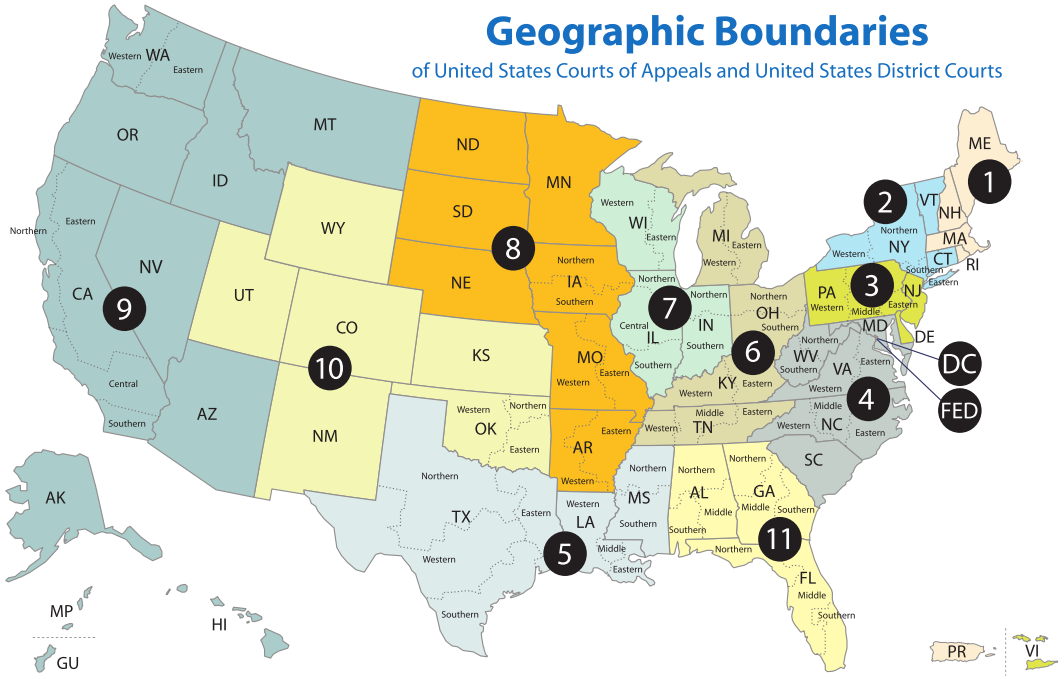
RegChallenger: A RAG Pipeline for Assessing the Legal Vulnerability of Federal Regulations

Henry Hirsch

Problem

- Agencies regulate nearly every sector of the economy and most aspects of Americans' daily lives. For example: EPA, FDA SEC, DHS, DOJ, etc.
- Agencies issue ~3,000-4,000 regulations (rules) per year in the Federal Register, the daily journal of the U.S. govt.
- Many rules are challenged in federal courts by regulated entities and advocates (companies, trade associations, advocacy groups, etc.)
- These challenges tend to occur in federal courts due to the Administrative Procedure Act, which grants U.S. federal courts oversight over all agency actions.
- Loper Bright (June 2024) overturned 40 years of Chevron deference. Courts now interpret statutes anew.
- No systematic tool to:
  - Assess which regulations are most vulnerable to legal challenge
  - Understand what legal characteristics make a rule vulnerable
  - Search for historical challenges to similar rules
- Litigation is a **time-consuming** and **expensive** process for both agencies and the groups challenging their rules in court.



Solution

- Compile data from Federal Judicial Center's Integrated Database, CourtListener's (Free Law Project Bulk Legal Data),and Federal Register's API to construct a retrieval-augmented generation (RAG) pipeline for assessing the legal vulnerability of federal regulations
- Three-step pipeline to join these data sources together and isolate 251 confirmed rulemaking challenges
- Then calculate historical analogs for a selected rule, rank according to legal relevance, and then use LLM prompts to weigh the ranked historical analogs as evidence for and against the selected rule and then make a vulnerability assessment (minimal, moderate, high).
  - Hybrid retrieval: semantic embedding + categorical boosts for agency, doctrine era, and administration (cosine similarity).
  - LLM reranker brings doctrinal fit (Chevron, Loper Bright, major questions) that pure embeddings miss.
  - Every prediction cites the specific historical analogs it relied on (**auditable, not a black box**).

Tech Stack

Data & Retrieval

- Python 3, pandas, numpy
- sentence-transformers
- all-mpnet-base-v2 (768-dim)
- Hybrid similarity scoring
- On-disk embeddings cache

AI & Reasoning

- Anthropic Claude API
- claude-sonnet-4 (temperature=0)
- Structured output prompts
- LLM reranker over embeddings
- Historical-outcome narratives

Interface & Data Sources

- Streamlit (custom CSS, GW theme)
- Plotly charts
- Federal Register API (live)
- CourtListener (opinion text)
- FJC Integrated Database (IDB)

~4,200 opinion files • 251 rulemaking reference cases • live FR search over ~470 agencies